

4.12.13

AI25BTECH11008 - Chiruvella Harshith Sharan

Question:

If the equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is $(2, -1)$, then find the length of the side of the triangle.

Answer:

Step 1: Represent the problem in vector form

The equation of the base of the triangle is

$$\mathbf{n}^T \mathbf{x} = 2 \quad (0.1)$$

where $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is the normal vector. The vertex is

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Step 2: Foot of perpendicular from vertex to base

The distance from $\mathbf{A}$ to the line is obtained by projecting $\mathbf{A}$ onto the normal:

$$d = \frac{\mathbf{n}^T \mathbf{A} - 2}{\|\mathbf{n}\|} = \frac{1 \cdot 2 + 1 \cdot (-1) - 2}{\sqrt{1^2 + 1^2}} = \frac{-1}{\sqrt{2}} \quad (0.2)$$

Hence, the foot of perpendicular $\mathbf{D}$ is:

$$\mathbf{D} = \mathbf{A} - d \frac{\mathbf{n}}{\|\mathbf{n}\|} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} - \left(-\frac{1}{\sqrt{2}}\right) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 + \frac{1}{2} \\ -1 + \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ -\frac{1}{2} \end{pmatrix} \quad (0.3)$$

Step 3: Length relation in equilateral triangle

In an equilateral triangle, the altitude length AD is related to the side a by:

$$AD = \frac{\sqrt{3}}{2} a \quad (0.4)$$

Here $AD = \|\mathbf{A} - \mathbf{D}\|$.

$$AD = \sqrt{\left(2 - \frac{5}{2}\right)^2 + \left(-1 + \frac{1}{2}\right)^2} = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$$

Step 4: Solve for side length

From (3):

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{2}a \implies a = \frac{2}{\sqrt{2}\sqrt{3}} = \sqrt{\frac{2}{3}} \quad (0.5)$$

Final Answer:

$$a = \sqrt{\frac{2}{3}}$$

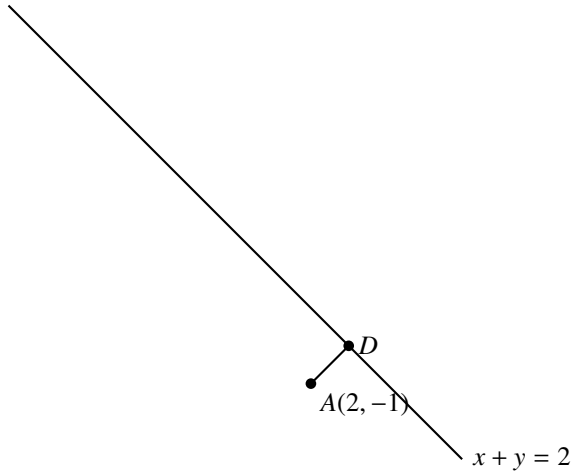


Fig. 0.1: Geometry of equilateral triangle with base $x + y = 2$